Centered Pell gaps and adaptive congruence moduli

Explicit improvements for Diophantine consistency equations

Research contribution prepared for Vladimir Reshetnikov
ProveIt project

23 October 2026

Abstract

We give two concrete improvements to the latest supplied ProveIt constructions. A fixed polynomial in thirteen strictly positive integer unknowns has exact degree 30 and a 93-operation arithmetic circuit, improving the previous degree 34 at the same circuit cost. A second fixed polynomial in twelve strictly positive integer unknowns has exact degree 110 and 100 operations, improving the preceding degree-118, 104-operation construction on both measures. Their lack of positive integer solutions is equivalent to Con(ZFC) by ordinary mathematical proofs formalizable in ZFC, composed with the pinned native compiler and refutation calibration.

The first improvement uses an already recovered divisibility condition to replace a fifth-power congruence modulus by a lower-degree adaptive modulus. The possible common divisor three is handled exactly. The second uses the small positive gap between two auxiliary Pell coordinates to merge the strong Pell root and auxiliary witness, and then centers the strong root before taking the norm. We prove that the exceptional zero denominator is impossible. The archive contains the four actual coefficient numerals, their reconstruction recipe, literal polynomial circuits, independent exact checks, and full proofs of the new reductions. These results do not lower the existing nine-positive-variable frontier or establish global minimality. They are specific improvements, not a claimed completely new consistency encoding.

Contents

1 Outcome, comparison, and scope **2**

2 Four fixed numerals, with no hidden parameter **3**

3 Common polynomial registers **3**

4 The thirteen-variable equation **4**

5 A centered two-root elimination **7**

6 The twelve-variable equation **8**

7 Exact degrees and circuit accounting **10**

8 Domains: positive, nonnegative, and unrestricted **11**

9 A tempting shortcut that fails **11**

10 Dependencies, verification, and integration **12**

1 Outcome, comparison, and scope

1.1 What is improved

The comparison snapshot is ProveIt commit `a2cd79b9120fc75f48dcbef6e38d0f8b18d64d49`, which places sources 60–67 beside the earlier packages. In particular, source 61 already gives the strong twelve- and thirteen-positive-variable points used as our direct baselines [1]. The older collection summary alone would give a misleadingly weak baseline.

Construction	Unknowns	Degree	Operations	Residuals
Source 61	13	34	93	4
Adaptive modulus, this report	13	**30**	**93**	4
Source 61, product form	12	118	104	3
Centered gap, this report	12	**110**	**100**	3
Source 61, sum of squares	12	172	103	3
Centered gap, sum of squares	12	**164**	**99**	3

The first degree reduction is $4/34 \approx 11.76\%$. The primary twelve-variable degree reduction is $8/118 \approx 6.78\%$, accompanied by four fewer binary operations. All operations count: addition, subtraction, multiplication, squaring, and multiplication by a fixed numeral each cost one gate. Fixed integer leaves and reuse of a previously computed register are free. This is an arithmetic-circuit measure, not bit complexity.

The current repository also contains nine-positive-variable constructions, ten-integer constructions, and much lower-degree equations with more variables. None is displaced merely by the table above. The best results of this iteration are at the twelve- and thirteen-positive-variable slices. In particular, source 64's 41-integer quartic and source 62's 35-integer sextic and 30-integer octic remain better low-degree integer choices [4, 5]. A proposed combination of those sources was checked and found to double-count an already eliminated register; no additional variable reduction is claimed. Source 62 also retains a cheaper thirteen-positive-variable point of degree 114 in 83 operations. Our thirteen-variable result improves degree at the fixed 93-operation source-61 baseline; it does not dominate every thirteen-variable tradeoff.

1.2 What "provably equivalent" means here

Fix the usual arithmetic proof predicate for finite ZFC refutations used by the repository, and write

$$\mathrm{Con}(\mathrm{ZFC}) \equiv \neg\exists p\ \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot).$$

The intended theorem for our primary polynomial P_{12} is

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

We establish the new reductions in ordinary mathematics, with explicit witness recovery in both directions. The native coefficient/compiler theorem and the fixed K215/refutation calibration are inherited mathematical inputs, identified in Section 10; no new end-to-end Lean, Rocq, or object-language ZFC certificate is supplied. Exact symbolic checks verify the finite algebra and literal circuits. They do not replace the unbounded compiler proof. Equation (1) asserts an equivalence, not either consistency or inconsistency.

The arithmetic construction uses the repository's native kernel. It does not use Jones's universal polynomial as its finalizer. Alternative source architectures were considered, but no complete

replacement with a better measured endpoint was obtained. No claim of literature priority is made for elementary norm or Pell identities; the new contribution is the precise combination and its verified effect on these fixed equations.

2 Four fixed numerals, with no hidden parameter

Every equation here uses the same complete source-44 compact tuple [2]:

$$(\beta, K_0, M_C, M_F), \qquad \beta = 2^{590069} - 1.$$

The archive contains their actual bytes in four gzip files. A decompressed byte string $b_0, \ldots, b_{m-1}$ denotes the integer $\sum_{j<m} b_j 256^j$. The port named K_0 is the payload named `K`; the port M_F is `MF_source`, which already includes the shift by β. One must use all four together.

Port	Bit length	Raw bytes	Payload
β	590069	73759	`beta.bin.gz`
K_0	592884	74111	`K.bin.gz`
M_C	590069	73759	`MC.bin.gz`
M_F	590070	73759	`MF_source.bin.gz`
Total		295388	

These are fixed positive integers, not extra quantified variables and not constants chosen according to the truth of consistency. The supplied `upstream/build_coefficients.py` reconstructs them from the supplied 430-row transition table, reconstructs the 943-tile language, and checks the finite packing certificate. Its generated raw numeral hashes are compared against the delivered binding by `verify.py`. This recipe is a finite computation, not an unbounded proof or halting search.

The numerical magnitude is still large. This iteration reduces arithmetic degree and circuit size, not coefficient height. Naming a large numeral is legitimate in a polynomial but should not be advertised as reducing its binary description length. The supplied payloads and recipe make that cost visible and reproducible.

3 Common polynomial registers

For the thirteen-variable construction supply precisely

$$j, F, \alpha, f, i, T, X, Y, r, \eta, \zeta, u, \gamma \in \mathbb{Z}_{>0}.$$

For the twelve-variable construction replace the pair (f, u) by one positive input n:

$$j, F, \alpha, i, T, X, Y, r, \eta, \zeta, n, \gamma \in \mathbb{Z}_{>0}.$$

All subsequent letters are polynomial abbreviations. They add no unknowns. For the thirteen-variable equation put $J = 3j + 1$; for the twelve-variable equation put $J = 6j + 1$. Define

$$q = \beta J + 1, \qquad C = q - F - \alpha, \qquad E = XY, \tag{2}$$

$$k = \eta + \zeta, \qquad c = kY + \eta, \qquad a = E + Y, \tag{3}$$

$$H = 4a + 3, \qquad \Delta = a^2 + H = (a+2)^2 - 1, \quad I = ic^2, \tag{4}$$

$$\ell = EkY, \qquad b_0 = X + ac, \tag{5}$$

$$R = \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)J. \tag{6}$$

The index defect, adaptive modulus, and its numerator are

$$r_0 = k - R - 1, \qquad L = qY, \qquad U = qL = q^2Y, \tag{7}$$

$$D_* = Ub_0 + H(U\gamma + LX - r_0). \tag{8}$$

For the first carrier put

$$W = (q-1)k + 1, \qquad M_0 = q^3(q-1), \tag{9}$$

$$B_t = (qK_0 + X)C - qF, \tag{10}$$

$$Z = M_0(\ell + k + r) + (q-1)Y - q^3 W B_t. \tag{11}$$

Here and below qF is multiplication by the supplied input F. The two carrier residuals are

$$A_0 = Z^2 - M_0^2(\ell(\ell+k)+1), \qquad A_1 = D_*^2 - U^2(1 + \Delta c^2). \tag{12}$$

Every power in these definitions is a fixed nonnegative integer power. There are no variable exponents, divisions, roots, digit functions, or implicit witnesses in the actual polynomials.

4 The thirteen-variable equation

With the common registers and $J = 3j + 1$, define

$$Q_f = \Delta(f^2 - 1), \qquad V_f = f^2(cT - R) - c, \tag{13}$$

$$A_s = f^2 - \Delta I^2 - 1, \tag{14}$$

$$A_a = (V_f + u)^2 - Q_f u(2V_f + u) - 1. \tag{15}$$

Our first equation is

$$\boxed{P_{13} = A_0^2 + A_1^2 + A_a^2 + A_s^2 = 0, \quad \mathbf{x} \in \mathbb{Z}_{>0}^{13}.} \tag{16}$$

Theorem 4.1. *Equation (16) has a positive integer solution exactly when the pinned native consistency kernel has one. Consequently its unsolvability is ZFC-provably equivalent to* Con(ZFC)*. Its exact total degree is* 30*, and the supplied literal evaluator has* 93 *live binary operations.*

We give the new argument in detail because a careless cancellation in its congruence step would invalidate the improvement.

4.1 Recover the first scale and the sign of the transport

At a positive zero all four residuals vanish. Since $M_0 > 0$, the equation $A_0 = 0$ says Z/M_0 is a rational square root of the integer $\ell(\ell+k)+1$. A rational square root of an integer is an integer. Thus $M_0 \mid Z$. Reducing this divisibility modulo q^3, and then modulo $q-1$, gives

$$q^3 \mid Y, \qquad q - 1 \mid B_t.$$

The reductions use $\gcd(q, q-1) = 1$ and $W \equiv 1 \pmod{q-1}$. Set $s = Y/q^3 > 0$, $t = B_t/(q-1)$, and $\tau = Z/M_0$. Then

$$\tau^2 = \ell(\ell+k)+1, \qquad \tau = \ell + k + r + s - Wt. \tag{17}$$

Because $k \geq 2$, one has $|\tau| < \ell + k$. Hence $Wt > r + s > 0$, so $t > 0$ and $B_t > 0$. But $C \leq 0$ would give $(qK_0 + X)C - qF < 0$. We have proved $C > 0$ before using any divisibility of X. The inherited compact-mask bounds now give

$$q \geq 16, \qquad 3q < R < q^4 - 2. \tag{18}$$

Since 590069 is odd, $\beta \equiv 1 \pmod 3$. Our restriction on J gives $q \equiv 2 \pmod 3$. Since $q \mid Y$, $H = 4Y(X+1) + 3 \equiv 3 \pmod q$, and therefore

$$\gcd(q, H) = 1. \tag{19}$$

4.2 Recover the main Pell index before the second scale

The numerator D_* is positive: $r_0 < k$, $c > kY$, and $U \geq q^5$ imply

$$D_* > Uac - Hk > k(UaY - H) > 0.$$

The last inequality follows immediately from $q \geq 16$, $Y \geq q^3$, and $H = 4a + 3$. Applying rational-square integrality to $A_1 = 0$ gives an integer $g_0 = D_*/U > 0$ with

$$g_0^2 - \Delta c^2 = 1, \qquad U \mid H(LX - r_0). \tag{20}$$

Write $A = a + 2$ and $P_0 = 2XY^2 + 1$. Use the usual Pell sequences

$$\chi_b(m) + \psi_b(m)\sqrt{b^2 - 1} = (b + \sqrt{b^2 - 1})^m.$$

The source Pell classification supplies positive indices p, n_0 with

$$c = \psi_A(p), \quad g_0 = \chi_A(p), \quad k = 2\psi_{P_0}(n_0), \quad |\tau| = \chi_{P_0}(n_0).$$

Here $c > 1$ excludes $p = 1$, because $\psi_A(1) = 1$. The case $n_0 = 1$ would give $k = 2$, $\eta = \zeta = 1$, and $c = 2Y + 1$, whereas $p \geq 2$ gives $c \geq 2A \geq 4Y + 4$. Therefore $n_0 \geq 2$, and

$$c > kY > 8XY^3 \geq 8q^9 > 2R, \qquad n_0 < p, \qquad c > 2p. \tag{21}$$

The comparison $n_0 < p$ uses $P_0 > A$ and $c > k$.

The normalized strong equation $A_s = 0$, with $I = ic^2$, gives $f = \chi_A(m)$, $I = \psi_A(m)$, $c \mid m$, and $m > 2p$. For clarity, strong divisibility first gives $p \mid m$; writing $m = pv$, the composition formula yields $I/c = \psi_{g_0}(v)$. Since $c \mid I/c$ and $g_0^2 \equiv 1 \pmod c$, its recurrence gives $\psi_{g_0}(v) \equiv v g_0^{v-1} \pmod c$. The residue g_0 is invertible modulo c, so $c \mid v$. Thus $m \geq pc > 2p$.

Now $Q_f = (\Delta I)^2$ and $A_a = 0$ yield

$$(\Delta I)^2 V_f^2 - ((\Delta I)^2 - 1)|V_f + u|^2 = 1.$$

Furthermore $V_f \equiv -c \pmod f$ and $V_f \equiv -R \pmod c$. Together with (21), these are precisely the hypotheses of the retained signed auxiliary-index lemma [1]. It gives $p = R$. A complete retained proof is supplied in `upstream/SIGNED_INDEX_LEMMA.md`. This argument has not assumed $q \mid X$ or reconstructed the missing index quotient.

4.3 The adaptive modulus and its common divisor three

The old source uses $U = q^5$ and $L = q^4$. We use $U = q^2Y$ and $L = qY$. This decreases their formal degrees from $5, 4$ to $3, 2$ while preserving the size needed for index recovery. The precise reason is the already proved divisibility $q^3 \mid Y$. Adaptive and successive congruence recovery already appear in sources 64 and 65 [5, 6]. The contribution here is their exact integration with the source-61 thirteen-variable carrier, using the modulus q^2Y and the following cofactor argument.

Put

$$h_3 = \gcd(H, U) = \gcd(H, Y) = \gcd(3, Y) \in \{1, 3\}, \qquad \mu = \frac{qY}{h_3}.$$

Since $\gcd(q, h_3) = 1$, dividing Y by h_3 preserves $q^3 \mid Y/h_3$. In particular

$$\mu \geq q^4 > R. \tag{22}$$

Canceling exactly h_3 in (20) shows $U/h_3 \mid LX - r_0$, and reduction modulo $\mu = L/h_3$ gives $\mu \mid r_0$. The first Pell recurrence gives

$$k \equiv 2n_0 \pmod{4XY^2},$$

and $\mu \mid 4XY^2$. Thus $\mu \mid 2n_0 - R - 1$. Since $1 \leq n_0 < p = R < \mu$, the absolute value is smaller than μ; therefore

$$2n_0 = R + 1. \tag{23}$$

It follows that $r_0 = k - 2n_0$ is a positive multiple of $4XY^2$. Because $q^2 \mid Y$, this is also a multiple of $U = q^2Y$. Equation (20) now gives $U \mid HLX$; as $U = qL$, this implies $q \mid HX$. By (19), $q \mid X$. The important distinction is that the common factor three is removed from an already divisible scale. Merely estimating $qY/3 \geq q^4/3$ would be too weak and would miss the actual argument.

4.4 Restore all deleted positive witnesses

Define

$$w = X/q, \qquad h = r_0/E, \qquad \gamma_{\mathrm{old}} = \gamma + w - r_0/U.$$

These are integers; the first two are positive. The main root satisfies $g_0 = X + ac + H\gamma_{\mathrm{old}}$. Since $g_0 > (a + 1)c$ and $c > X$, it follows that $H\gamma_{\mathrm{old}} > c - X > 0$, so the third is positive as well.

Now $B_t = q[(K_0 + w)C - F]$. Its positivity and its divisibility by $q - 1$ give a positive integer z_{new} with

$$(K_0 + w)C - F = (q - 1)z_{\mathrm{new}}.$$

Set $z_{\mathrm{old}} = z_{\mathrm{new}} + 1$; then

$$(K_0 + w)C + (q - F) - z_{\mathrm{old}}(q - 1) = 1.$$

With $T_{\mathrm{old}} = fT$, the roots $|\tau|, |V_f + u|$, and the restored index quotient h, this is the full retained normalized source interface. To use the ordinary strong-equation interface instead, put $i_{\mathrm{old}} = \Delta i$; then $(i_{\mathrm{old}}c^2)^2 = \Delta(f^2 - 1)$. Source soundness gives a finite ZFC refutation.

4.5 Construct positive new witnesses from a refutation

A finite refutation supplies the canonical compact witness of the inherited compiler. It has $q = B^N$, $B = 2^{590069}$, and $J = 1 + B + \cdots + B^{N-1}$ for odd $N > 1$. Hence $J \equiv 1 \pmod 3$, $j = (J - 1)/3 > 0$, and J is odd. Its other properties include

$$X = 2^R, \quad w = X/q > q, \quad Y = sq^3, \quad R \equiv 3 \pmod 4, \quad \gamma_{\mathrm{old}} > w.$$

The retained gap estimates and auxiliary construction are documented in source 58 and source 61 [3, 1]. In particular r_0 is a positive multiple of $4XY^2$, so

$$\gamma = \gamma_{\mathrm{old}} - w + r_0/U > 0$$

is integral and makes $D_* = Ug_0$. For the first carrier choose

$$t = q(z_{\mathrm{old}} - 1), \qquad r = Wt + \tau - \ell - k - s.$$

Here $z_{\mathrm{old}} \geq 2$ canonically, $t \geq q$, and $\tau > \ell$, $k > s$, so $r > qW - k - s > 0$. Direct substitution gives $Z = M_0\tau$. Finally reselect the canonical squared-modulus auxiliary pair, retaining positive f, i, T, u. This gives $A_s = A_a = 0$. All supplied coordinates are positive, completing the converse. No transformation of an arbitrary old witness is claimed; canonical witness reselection is sufficient for the existence equivalence.

5 A centered two-root elimination

The following local lemma is the additional ingredient for twelve variables. It applies before any compiler reasoning and is useful independently. The two-root norm identity and the strategy of centering a root are already used in source 62, Sections 2–3 of its positive-thirteen proof [4]. That construction centers the first Pell root at ℓ. Here we merge the strong root with the auxiliary gap and center it at $(a+2)I$. The distinct center, its positivity bound, and the exceptional-case exclusion produce the twelve-variable improvement below.

Lemma 5.1 (Centered Pell-gap norm)**.** *Let $a \geq 2$, $c \geq 3$, and $i \geq 1$ be integers, and let b be any integer. Put*

$$\Delta = (a+2)^2 - 1, \qquad I = ic^2, \qquad D = 1 + \Delta I^2, \tag{24}$$

$$Q = \Delta^2 I^2, \qquad d = Q - 1, \qquad f_0 = (a+2)I, \tag{25}$$

$$e = 1 - I^2, \qquad V = Db - c. \tag{26}$$

For an integer n define

$$x = n + f_0 + V, \qquad E_a = QV^2 - 1, \tag{27}$$

$$N = \left(d(x^2 - D) - E_a\right)^2 - 4dDE_a. \tag{28}$$

If $N = 0$, there exist positive integers f, u with

$$f^2 = D, \qquad QV^2 - d(V+u)^2 = 1. \tag{29}$$

Conversely, if a positive solution of (29) satisfies $u > f_0$, then $n = f - f_0 + u$ is positive and makes $N = 0$.

Proof. The centered error is exactly $D - f_0^2 = e = 1 - I^2$. Define the proof registers

$$B_n = 2dx, \qquad H_n = d(n^2 + 2Vn + e) - V^2 + 1.$$

Two elementary identities give

$$N = \left(d(x^2 + D) - E_a\right)^2 - B_n^2 D, \tag{30}$$

$$d(x^2 + D) - E_a = H_n + B_n f_0. \tag{31}$$

Consequently

$$N = (H_n + B_n f_0)^2 - B_n^2 D = H_n^2 + 2B_n f_0 H_n - B_n^2 e. \tag{32}$$

We must exclude $B_n = 0$, rather than divide without justification. Since $d > 0$, that case would give $x = 0$ and

$$\begin{aligned} H_n &= dD - QV^2 + 1 \\ &= Q(D - V^2) - D + 1 \\ &= \Delta I^2(\Delta(D - V^2) - 1) \neq 0. \end{aligned}$$

The last inequality uses $\Delta > 1$: no integer $D - V^2$ can have $\Delta(D - V^2) = 1$. Hence $N = H_n^2 \neq 0$, a contradiction.

For $N = 0$, therefore, $f_* = (H_n + B_n f_0)/B_n$ is a rational square root of the integer D. It is an integer. Put $y_* = x - f_*$. Expanding the preceding identities gives

$$QV^2 - dy_*^2 = 1.$$

Also $|V| > 1$. Indeed $D > c + 1$; for $b \leq 0$, $V \leq -c$, and for $b \geq 1$, $V \geq D - c > 1$. Therefore

$$d(y_*^2 - V^2) = V^2 - 1 > 0,$$

so $|y_*| > |V|$. The positive integers $f = |f_*|$ and $u = |y_*| - V$ satisfy (29).

Conversely put $n = f - f_0 + u$. Then $x = f + (V + u)$, and substituting these two roots into the uncentered norm gives $N = 0$. The stated gap bound ensures $n > 0$. □

Remark 5.2. The soundness direction does not require $n > 0$ and allows both signs of the intermediate roots. Positivity is recovered afterwards. The converse does require room for the subtraction of f_0; this is supplied by the following canonical Pell estimate, not by an unchecked change of variables.

Lemma 5.3 (Canonical gap)**.** *For $M \geq 2$ and odd $R \geq 5$,*

$$\psi_M(R) - \frac{\chi_M(R)}{M} = \frac{\psi_M(R-1)}{M} \geq 8M^2 - 4.$$

Both terms on the left are integers.

Proof. The recurrence identity $\chi_M(R) = M\psi_M(R) - \psi_M(R-1)$ proves the equality. For odd R, $\chi_M(R)$ is a multiple of M by the odd-index Chebyshev recurrence. The sequence $\psi_M(t)$ increases for $t \geq 0$; $R - 1 \geq 4$ and $\psi_M(4) = 8M^3 - 4M$ give the bound. □

The proof uses a very large canonical auxiliary gap to pay for the small polynomial center. It does not estimate the size of a ZFC refutation or assume that one exists.

6 The twelve-variable equation

Use the twelve positive inputs from Section 3, now with $J = 6j + 1$. Retain A_0, A_1 from (12). Put

$$D = 1 + \Delta I^2, \qquad Q = \Delta(D-1) = \Delta^2 I^2, \qquad d = Q - 1, \tag{33}$$

$$f_0 = (a+2)I, \qquad V = D(cT - R) - c, \qquad x = n + f_0 + V, \tag{34}$$

$$E_a = QV^2 - 1, \qquad N = \big(d(x^2 - D) - E_a\big)^2 - 4dDE_a. \tag{35}$$

The primary equation is

$$\boxed{P_{12} = (1 + A_0)(1 + A_1)(1 + N) - 1 = 0, \quad \mathbf{x} \in \mathbb{Z}_{>0}^{12}.} \tag{36}$$

This is a single fixed integer polynomial, fully defined by the displayed registers and the four actual numerals in Section 2.

Theorem 6.1. *Equation (36) has a positive integer zero if and only if there is a finite ZFC refutation. The proof is formalizable in ZFC using the retained calibration. The equation has exact total degree* 110 *and a* 100*-operation literal circuit.*

6.1 The product finalizer is exact

Because J and β are odd, q is even. Thus M_0 and U are even, and

$$A_0 \equiv Z^2 \pmod 4, \qquad A_1 \equiv D_*^2 \pmod 4.$$

Also $N \equiv \big(d(x^2 - D) - E_a\big)^2 \pmod 4$ directly from its definition. If $P_{12} = 0$, its three integer factors multiply to 1, so each is $+1$ or -1. A factor -1 would require its residual to be $-2 \equiv 2 \pmod 4$, which is not a square. All factors are therefore $+1$:

$$P_{12} = 0 \quad \Longleftrightarrow \quad A_0 = A_1 = N = 0. \tag{37}$$

No nonnegativity of the residuals is assumed. For a more familiar but higher-degree finalizer, the delivered companion is $A_0^2 + A_1^2 + N^2 = 0$.

6.2 Soundness and completeness

Given a positive zero of P_{12}, the local norm lemma with $b = cT - R$ restores positive f, u satisfying the strong and auxiliary residuals. The outer registers are unchanged. Set $j_{13} = 2j$; then $3j_{13} + 1 = 6j + 1$, so the recovered tuple is a zero of the proved thirteen-variable polynomial. Its soundness gives a finite ZFC refutation.

Conversely take the canonical source tuple for a refutation. Its repunit J is odd and 1 modulo 3, hence 1 modulo 6; padding gives $J > 1$. Thus $(J-1)/6$ is a positive integer. Use the adaptive-carrier construction above. For the normalized strong witness set $M = \Delta I$. The canonical auxiliary pair is

$$V = \chi_M(R)/M, \qquad y = \psi_M(R),$$

with $R \equiv 3 \pmod 4$ and $R > 3q \geq 48$. The squared-modulus source construction supplies positive integer T with $V = D(cT - R) - c$. The canonical gap lemma gives

$$u = y - V \geq 8M^2 - 4 > f_0,$$

because $f_0 = (a + 2)I < \Delta I = M$. Hence

$$n = f - f_0 + u > 0$$

is an admissible new input. The norm lemma gives $N = 0$, and the first two residuals already vanish. This proves the converse and therefore Equation (1).

7 Exact degrees and circuit accounting

All supplied unknowns have degree one and the four fixed numerals degree zero. The actual degree is often lower than naive degree propagation through the evaluation graph. The verifiers check the cancellations rather than reporting such propagation as an exact result.

Put $q_* = 3\beta j$ or $6\beta j$ as appropriate, and write

$$k_* = \eta + \zeta, \quad a_* = XY, \quad c_* = k_* Y, \quad \Delta_* = X^2 Y^2, \quad I_* = i k_*^2 Y^2,$$

$$R_* = q_*^3 (q_* - F), \qquad B_{t,*} = (K_0 q_* + X)(q_* - F - \alpha) - q_* F.$$

The carrier leading forms, checked by full exact expansion of the low-degree residuals, are

$$\mathrm{in}_{15}(A_0) = -2 q_*^8 k_*^2 X Y^2 B_{t,*}, \tag{38}$$

$$\mathrm{in}_{13}(A_1) = 8 q_*^2 k_* X^2 Y^4 (q_*^2 Y \gamma + R_*). \tag{39}$$

In particular the sign before R_* in (39) is positive. The other thirteen-variable residual degrees are 13 and 14. Only A_0^2 reaches degree 30, giving

$$\mathrm{in}_{30}(P_{13}) = 4 q_*^{16} k_*^4 X^2 Y^4 B_{t,*}^2.$$

The coefficient of $j^{16}\eta^4 X^4 Y^4 \alpha^2$ is $4(3\beta)^{16} > 0$. Specializing the actual fixed numerals therefore cannot cancel this top degree.

For the centered norm, the convenient degree certificate is Equation (32), not the shorter literal evaluator. The degrees are

Register	D	Q, d	V	f_0	e	B_n	H_n	N
Degree	14	18	18	7	10	36	37	82

The three terms H_n^2, $2B_n f_0 H_n$, and $-B_n^2 e$ have degrees at most $74, 80, 82$. The last is uniquely highest, and

$$\mathrm{in}_{82}(N) = 4\Delta_*^6 I_*^{10} R_*^2 \neq 0. \tag{40}$$

Thus the product finalizer has exact degree $15 + 13 + 82 = 110$, and the sum-of-squares companion has degree 164. A specific nonzero coefficient in P_{12} is

$$[j^{18} F^2 \alpha \gamma i^{10} X^{16} Y^{39} \eta^{23}] P_{12} = 64(6\beta)^{18} > 0. \tag{41}$$

Its total exponent is 110. The analogous top coefficient in the sum-of-squares companion is

$$[j^{12} F^4 i^{20} X^{24} Y^{64} \eta^{40}] P_{12,\mathrm{sos}} = 16(6\beta)^{12} > 0.$$

The literal primary circuit evaluates N by the shorter Equation (28). The proof certificate uses the algebraically equal centered expression to expose degree cancellation. Neither expression introduces an additional witness. The graph is the exact polynomial definition; its registers are ordered abbreviations, and every gate and input is live.

8 Domains: positive, nonnegative, and unrestricted

The small variable counts above are over $\mathbb{Z}_{>0}$. They must not be relabeled as unrestricted integer counts. For nonnegative integers simply replace each of the twelve inputs by $x + 1$. The delivered `circuits/natural12.json` has degree 110 and 112 gates.

For an entirely unrestricted integer equation, define

$$\theta(a, b, c) = 1 + a^2 + b^2 + c(c + 1).$$

This takes exactly the positive integer values. Nonnegativity is immediate. For surjectivity, Legendre's three-square theorem applied to $4m + 1$ gives one odd and two even squares; write

$$4m + 1 = (2a)^2 + (2b)^2 + (2c + 1)^2,$$

so $m = a^2 + b^2 + c(c + 1)$. This familiar conversion is also recorded in Lemma 6.1 of Bayer et al. [7].

Applying θ to each input gives a fully specified equation in 36 unrestricted integers, of exact degree 220 and 184 gates. Its unsolvability has the same ZFC equivalence. It is included to remove any domain ambiguity, not as a new integer frontier point. Its degree is exact: the homogeneous leading substitution maps each original variable to a sum of three distinct squares; setting the second and third members of each triple to zero reduces to the injective substitution $x_i \mapsto a_i^2$.

9 A tempting shortcut that fails

One might try to replace $X = wq$, $Y = sq^3$ by $X = wq^2$, $Y = sq^2$, preserving $XY \geq q^4$ while lowering the first carrier's degree. The compiler does not depend only on the product XY.

For $q = 2^t$, its half-binomial encoding has a packed index

$$R = (q^2 - S)(q^2 - 1) + T, \qquad \mathrm{pc}(T) = t + 2.$$

The relevant inverse-population lemma gives $\mathrm{pc}(R) \leq 3t+2$, with equality exactly when the required masks vanish. The corresponding ordinate valuation is $\mathrm{pc}(R) - 2$. Thus $q^3 \mid Y$ enforces the full threshold, whereas $q^2 \mid Y$ does not. This role of the cubic scale is explicit in the retained source notes, cited with exact blobs in `research/SCALE_BARRIER.md`.

An exact local negative control is

$$q = 16, \quad t = 4, \quad S = 32, \quad T = 175, \quad R = 57295.$$

Here $\mathrm{pc}(T) = 6 = t + 2$, $R \equiv 3 \pmod 4$, but $\mathrm{pc}(R) = 13 < 14 = 3t + 2$ and $S \,\&\, T = 32 \neq 0$. The weakened threshold $2t + 2 = 10$ accepts this invalid local mask configuration. This is a counterexample to the weakened packing predicate, not a purported zero of the full consistency polynomial. Moving a factor of q to X cannot repair this missing population condition by a size estimate alone.

A substantial further reduction will need a different local packing predicate, a lower-cost way to enforce that exact threshold, or a source formalism whose complete arithmetic compilation is smaller. Merely citing a smaller universal machine or the DPRM theorem does not supply the missing fixed polynomial or its coefficient binding.

10 Dependencies, verification, and integration

10.1 The proof boundary

The logical chain is

$$\begin{aligned} &\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12} : P_{12} = 0 \\ \iff &\text{ the normalized native kernel has a positive witness} \\ \iff &\exists p \ \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot). \end{aligned}$$

The first equivalence is the new work: adaptive congruence recovery, centered-root elimination, exceptional-case exclusion, positivity, and canonical reconstruction. The second is the explicitly retained compiler and calibration. No finite computation establishes that second equivalence by searching for all possible roots.

Input or verification	Exact role
Source 44 compact proof	Whole-tuple native compiler adapter, mask bounds, canonical padding and finite coefficient recipe. Retained verbatim.
Inherited ordinary proof and Pell audit	Native consistency kernel, power/half-binomial recovery, canonical normalized witnesses, and named K215/refutation calibration. Retained with source pins.
Sources 58 and 61	Squared-modulus auxiliary reselection, positive gaps, transport carrier, and signed index lemma. Retained and explicitly used.
New adaptive-carrier proof	Replaces q^5, q^4 by q^2Y, qY; handles the factor three and proves the same existence equivalence.
New centered-gap proof	Merges f, u into n; proves integrality, positivity, the absence of the exceptional branch, and the converse.
Exact finite checkers	Polynomial identities, leading forms, degree witnesses, gate liveness/counts, whole coefficient hashes and reconstruction.
Formalization status	No newly compiled end-to-end Lean/Rocq proof of the changed literal polynomial is claimed.

The primary new checks use only the Python standard library. They expand the low-degree carrier residuals in all witnesses and symbolic ports, and verify the large norm through exact identities in independent abstract registers. This avoids an unnecessary full expansion of the degree-110 polynomial while still proving the relevant polynomial identity. The nonzero coefficients in Section 7 certify degree after specialization to the actual huge fixed numerals.

Separate small modular and bounded tests are diagnostics for transcription and the factor-three argument. They are labeled as such. They are not evidence of consistency, semantic completeness, or global minimality. Independent checks in this archive were conducted by separate AI-assisted workers in this research session; they are not external human peer review.

10.2 Archive layout and reproducibility

The main commands are

```
python3 verify.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

- `report.tex`, `report.pdf`: this self-contained article.
- `build.py`, `build_domains.py`: regenerate the twelve-variable primary, sum-of-squares, and domain companion circuits.
- `positive13/`: generator, literal circuit, full proof, and independent exact checker for the thirteen-variable reduction.
- `circuits/`: authoritative division-free graphs for the twelve-variable forms and standard domain companions.
- `coefficients/`: four actual numeral payloads and the complete unchanged compact44 binding.
- `evaluate.py`: evaluate any delivered circuit at supplied integers, optionally modulo a chosen integer; no zero search.
- `checks/`, `VERIFICATION.json`: finite verification scripts and receipts stating their exact scope.
- `upstream/`: pinned proof dependencies, coefficient generator, literal transition table, and license/attribution records.
- `research/`: independent mathematical audit, rejected scale shortcut, and the low-degree exploration with its corrected counts.

The new files can be placed as a contribution beside the existing sources. The current smaller-variable and lower-operation alternatives should remain available. This report establishes two new measured points and their proof obligations; it does not identify a unique simplest equation across all domains, degrees, circuit sizes, and coefficient heights.

References

[1] V. Reshetnikov et al., ProveIt, source 61, *Congruence carriers*, snapshot `a2cd79b9120`, 23 October 2026. Pinned source directory. The proof and baseline README are retained under `upstream/`.

[2] ProveIt, source 44, *Complement-packed native clauses: a smaller complete coefficient tuple*, same snapshot. Pinned native tuple and proof.

[3] ProveIt, source 58, *Integral root*, same snapshot. Pinned source directory. The report source is retained as `upstream/SOURCE58_REPORT.tex`.

[4] ProveIt, source 62, *Sign and norm*, same snapshot. Pinned source directory.

[5] ProveIt, source 64, *Structural reductions*, same snapshot. Pinned source directory.

[6] ProveIt, source 65, *Shared taps*, same snapshot. Pinned source directory. Its Pell component uses successive congruence recovery.

[7] J. Bayer, M. David, M. Haßler, Y. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909, v2, 30 June 2025, Lemma 6.1. `https://arxiv.org/html/2506.20909v2`. The latest version observed during this work is v3, 22 October 2026; the cited elementary domain lemma is pinned to v2.